

Square Roots — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

$\sqrt{625} = ?$

- (a) 15 (b) 25 (c) 35 (d) 125

Q2 • MCQ**[1]**

By prime factorisation, $\sqrt{1156} = ?$

- (a) 34 (b) 36 (c) 32 (d) 44

Q3 • FILL IN THE BLANK**[1]**

The smallest number by which 2800 must be multiplied to become a perfect square is:

Q4 • MCQ**[1]**

Which of these numbers is a perfect square?

- (a) 2800 (b) 156 (c) 11025 (d) 250

Q5 • TRUE / FALSE**[1]**

Every perfect square has two integer square roots — one positive and one negative.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): 156 is a perfect square.

Reason (R): In $156 = 2 \times 2 \times 3 \times 13$, the prime factors cannot all be paired.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Using prime factorisation, find $\sqrt{4761}$.

Q8

[2]

Estimate $\sqrt{250}$ by stating the two consecutive whole numbers it lies between, and the nearer one.

Q9

[2]

Is 2800 a perfect square? Justify your answer using prime factorisation.

Long answer (3 marks each)

Q10

[3]

Using prime factorisation, find $\sqrt{11025}$. Show the pairing clearly.

Q11

[3]

By what smallest number should 216 be divided so that the quotient is a perfect square? Also find the square root of the quotient.

Q12

[3]

A square hall has an area of 1156 square metres. Find the length of one side, showing your method.

Total: 21 marks · Check your work against the [answer key](#).